

2U Server Conjugate Heat Transfer Simulation Report

OpenFOAM v2012 · chtMultiRegionSimpleFoam · 16-Core Parallel

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Parameter	Value
Solver	chtMultiRegionSimpleFoam
Mesh cells	~1.1 million
Parallel cores	16
Total iterations	20,000
CPU heat dissipation	150 W
Fan inlet velocity	2.86 m/s (Delta FFB0812VH)
Max CPU temperature	96°C
Inlet air temperature	27°C (300 K)

1. Project Overview

This project presents a steady-state Conjugate Heat Transfer (CHT) simulation of a 2U rack-mount server chassis using OpenFOAM v2012. The simulation couples fluid flow in the air domain with heat conduction through solid components including a CPU die, aluminium heatsink, and steel front panel. The goal is to predict thermal performance under a 150 W CPU load with forced air cooling from an 80 mm Delta fan.

The case demonstrates industrial-grade CFD methodology including multi-region mesh generation with snappyHexMesh, turbulent flow modelling with the k-epsilon model, conjugate heat transfer boundary conditions at solid-fluid interfaces, and large-scale parallel computation across 16 cores over 20,000 iterations.

Key Features

- Multi-region CHT with 4 coupled regions: fluid air domain, CPU silicon, aluminium heatsink, steel front panel
- Geometry built in Ansys SpaceClaim with STL export via scripted API
- Automated mesh pipeline: surfaceTransformPoints → snappyHexMesh → splitMeshRegions
- Fan modelled as a fixed velocity inlet (2.86 m/s) from Delta FFB0812VH fan curve
- TIM and PCB modelled as thermal contact resistance boundary conditions
- 16-core MPI parallel run on a 32-core workstation (hostname: rinkoa)
- Post-processing in ParaView with temperature, velocity, and pressure field visualisation

2. Geometry and Mesh

2.1 Server Geometry

The server geometry was constructed in Ansys SpaceClaim using the Python scripting API (V261). The model includes representative server components: CPU die (45×45×4 mm, silicon), thermal interface material (0.1 mm, $k = 6 \text{ W/m}\cdot\text{K}$), aluminium heatsink (120×80×40 mm with 15 fins), steel front panel with fan cutouts, motherboard, PCBs, and expansion cards. STL files were exported with FilePerBody granularity and scaled from millimetres to metres using surfaceTransformPoints.

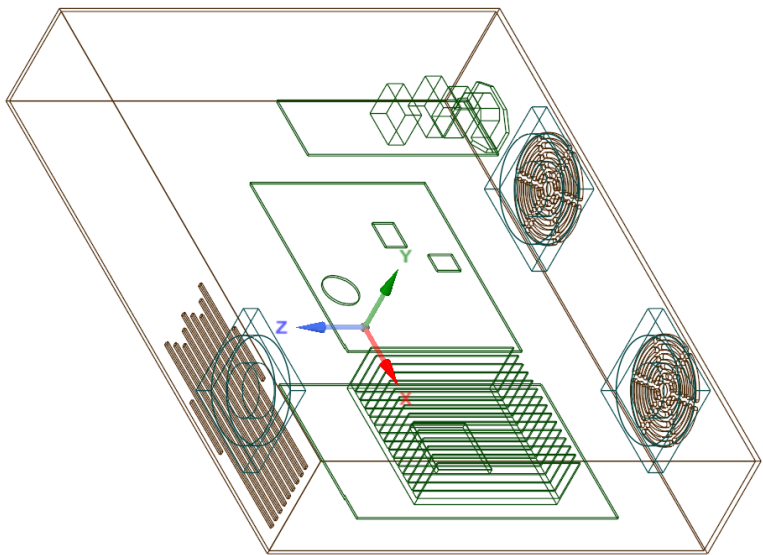


Figure 1: SpaceClaim geometry showing server chassis (orange), motherboard and PCBs (green), heatsink fin array (brown), and fan assemblies (blue).

2.2 Mesh Generation

The computational mesh was generated using OpenFOAM's snappyHexMesh utility with locationsInMesh to capture all regions. The background hex mesh was created with blockMesh at a base resolution sufficient to resolve fin channels. Refinement levels were applied to heatsink fins (level 3), CPU surfaces (level 4), and near-wall regions. The mesh was then split into regions using splitMeshRegions -cellZones, producing approximately 1.1 million cells across all four regions.

Region	Material	Approx. Cells	Role
domain0	Air (perfect gas)	~900,000	Fluid flow & heat transport
CPU	Silicon ($k=130 \text{ W/m}\cdot\text{K}$)	~5,200	Heat source (150 W)
Heatsink	Aluminium ($k=202 \text{ W/m}\cdot\text{K}$)	~150,000	Heat dissipation
FRONT_PANEL	Steel ($k=16 \text{ W/m}\cdot\text{K}$)	~50,000	Chassis thermal boundary

3. Simulation Setup

3.1 Solver Configuration

The steady-state CHT solver `chtMultiRegionSimpleFoam` was selected for its robustness and efficiency for coupled solid-fluid thermal problems. The fluid region uses the `heRhoThermo` thermophysical model with `perfectGas` equation of state, enabling buoyancy-driven flow effects through density variation with temperature. Turbulence is modelled using the standard `k-epsilon` RAS model.

3.2 Boundary Conditions

Boundary	Field	Type	Value
Fan_front_inlet	U	fixedValue	(0 0 -2.86) m/s
Fan_front_inlet	T	fixedValue	300 K (27°C)
Fan_front_outlet	U	inletOutlet	Zero gradient
Fan_front_outlet	p_rgh	fixedValue	0 Pa (gauge)
Walls	U	noSlip	-
Walls	T	zeroGradient	Adiabatic
CPU/domain0 interface	T	turbulentTemperatureCoupledBaffleMixed	Coupled
CPU heat source	h	scalarSemImplicitSource	150 W (1.863e7 W/m³)

3.3 Numerical Schemes

Spatial discretisation uses second-order Gauss linear schemes for divergence terms and Gauss linear corrected for Laplacian terms. The SIMPLE algorithm handles pressure-velocity coupling in the fluid domain. Relaxation factors of 0.7 for momentum, 0.5 for energy, and 1.0 for density ensure stable convergence. The simulation was run with `steadyState` `ddt` schemes and convergence residual targets of $1e-4$ for all fields.

4. Results

4.1 Convergence History

The simulation was run in stages across multiple restarts, accumulating 20,000 steady-state iterations. The CPU maximum temperature history shown below captures the full thermal development from the initial isothermal flow field (27°C) through to near-converged steady state. Two step changes in the curve correspond to solver restarts where coupled boundary conditions were re-established. The curve shows asymptotic behaviour approaching 96°C, indicating near-convergence.

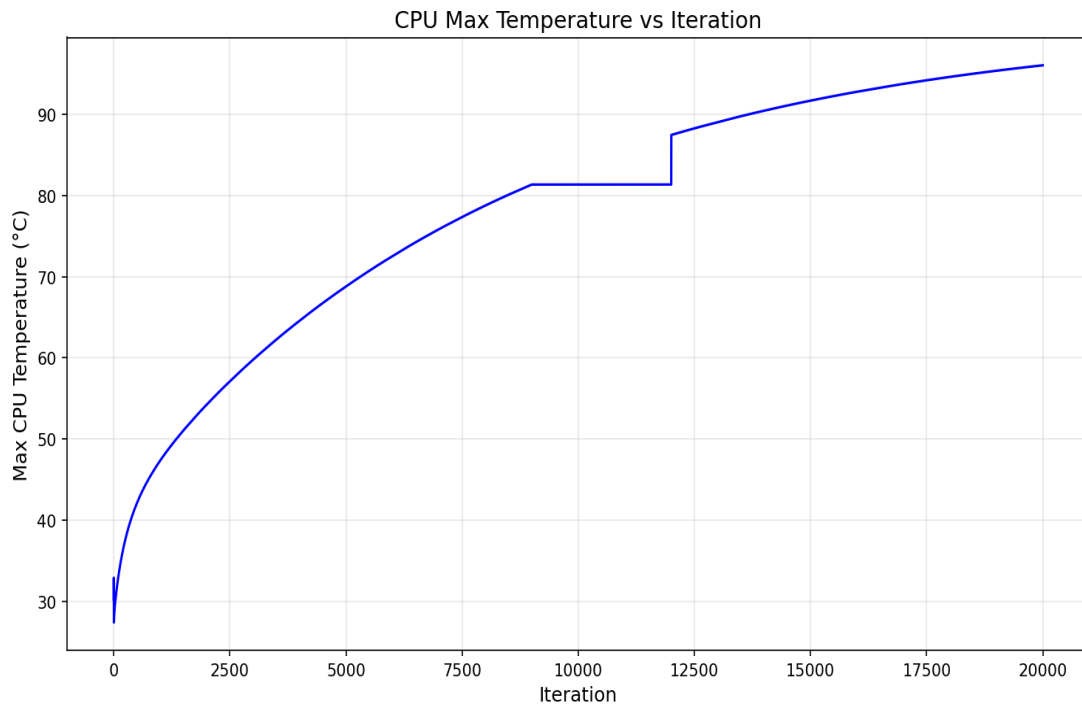


Figure 2: CPU maximum temperature vs. iteration number across all 20,000 steady-state iterations.

4.2 Temperature Distribution — Fluid Domain

The mid-plane temperature contour (Figure 3) shows the thermal plume rising from the heatsink base through the fin channels. Inlet air enters at 27°C and is heated as it passes through the fin array, exiting at significantly elevated temperatures. The hot spot is localised directly above the CPU die location, with the heatsink effectively spreading the heat laterally across the fin array.

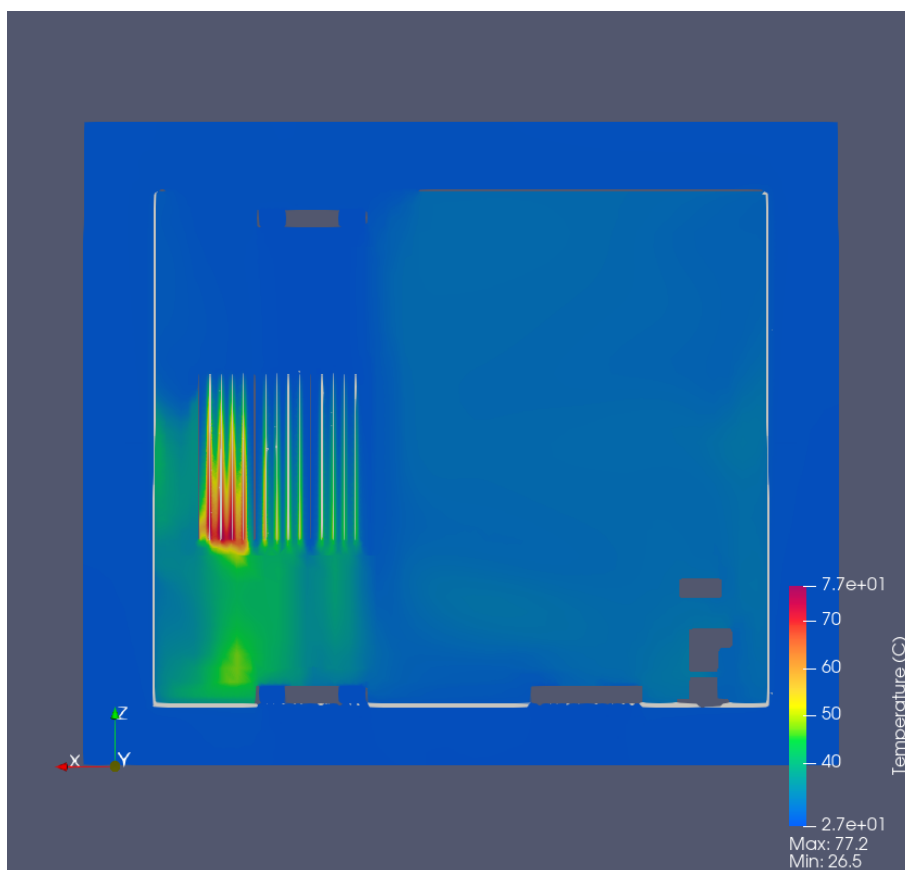


Figure 3: Temperature contour on vertical mid-plane slice through the server interior (°C).

4.3 Heatsink Thermal Plume

Figure 3 shows a side-view cross-section through the heatsink region, revealing the thermal plume structure. The CPU die is visible at the base (red, $\sim 96^{\circ}\text{C}$) with heat conducted upward through the heatsink base and into the fins. The plume extends downstream in the flow direction as heated air exits the fin channels, reaching peak air temperatures of $\sim 88^{\circ}\text{C}$ immediately downstream of the hottest fin sections.

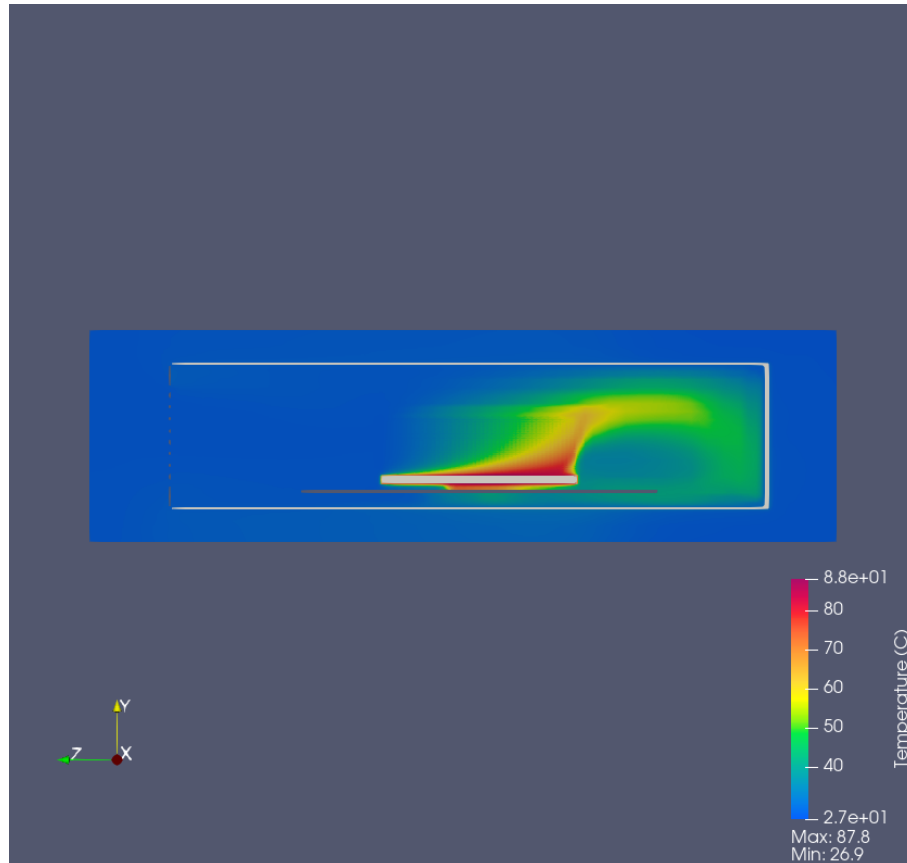


Figure 4: Side-view temperature contour showing thermal plume through the heatsink fin array ($^{\circ}\text{C}$).

4.4 Solid Region Temperatures

Figure 4 presents the temperature distribution across all solid regions. The CPU die reaches a maximum junction temperature of 96°C under 150 W dissipation. Heat is conducted through the TIM layer (modelled as contact resistance) into the heatsink base, spreading laterally across the 120×80 mm base plate before conducting into the 15 aluminium fins. The front panel reaches only $\sim 28^{\circ}\text{C}$ due to its steel construction and distance from the heat source.

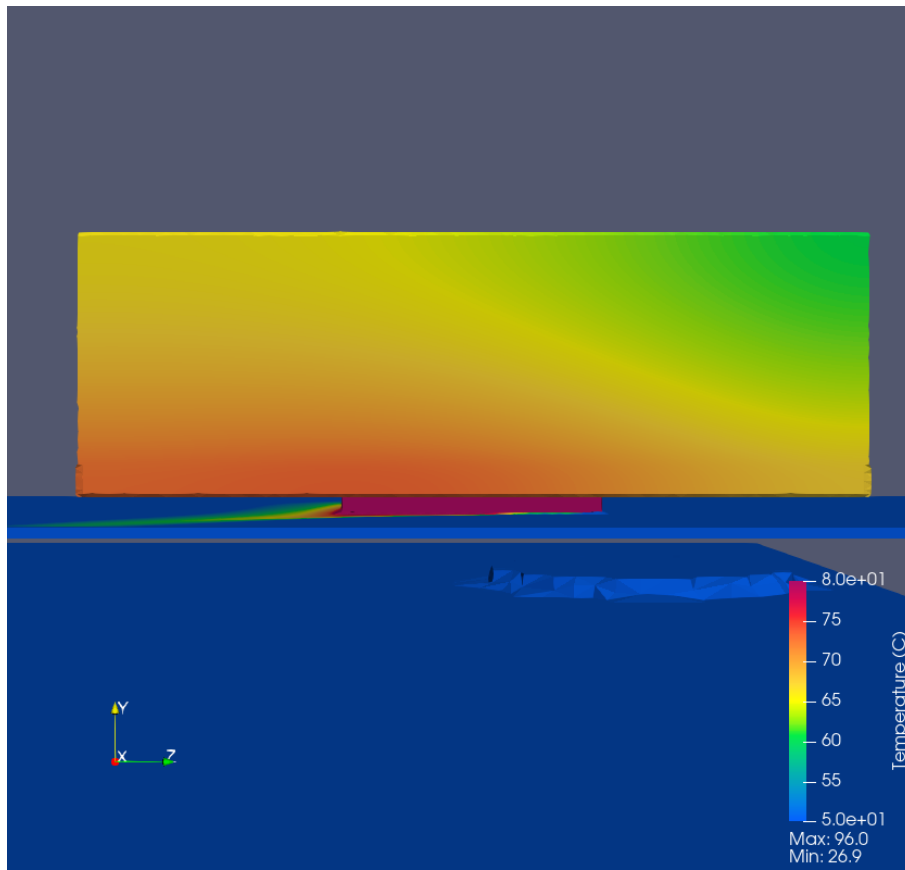


Figure 5: Temperature distribution across solid regions — CPU (top), motherboard PCB (middle), heatsink with fins (bottom).

4.5 Velocity Distribution

The velocity magnitude contour (Figure 5) shows the airflow field within the server. Peak velocities of ~11 m/s are observed within the heatsink fin channels due to flow acceleration through the reduced cross-sectional area. The fan inlet jets are clearly visible at the front of the chassis, with jets impinging on the motherboard and PCB components before being redirected toward the heatsink. Recirculation zones are present in the wake of large board-mounted components.

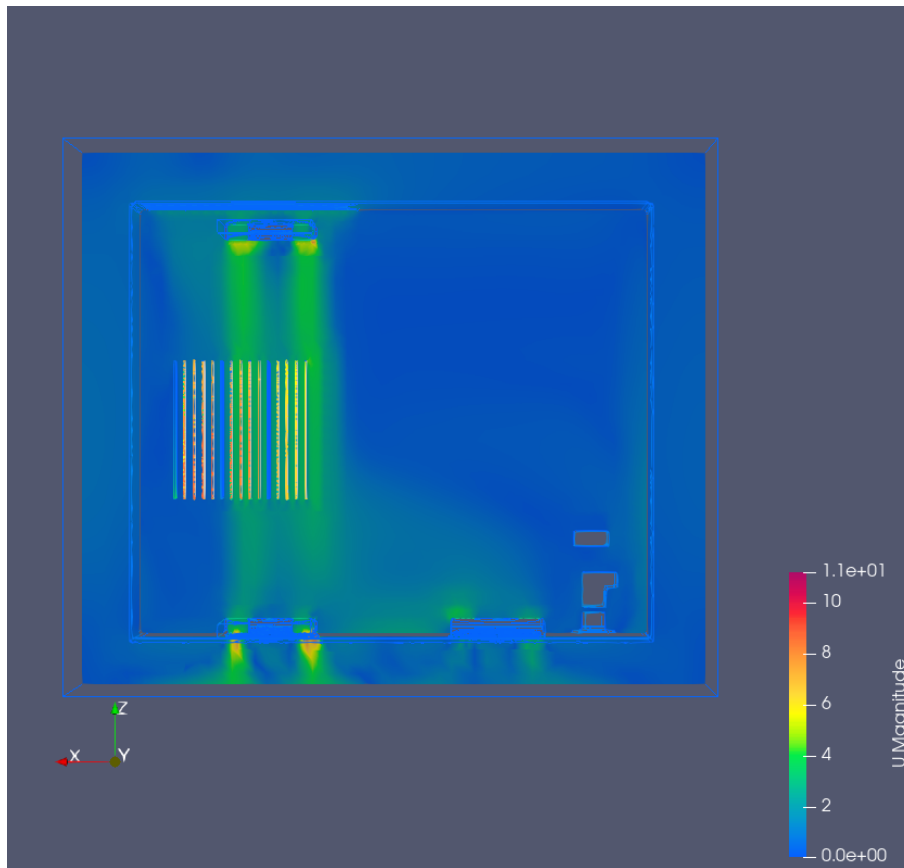


Figure 6: Velocity magnitude contour on mid-plane slice showing airflow distribution (m/s).

4.6 Pressure Distribution

Figure 6 shows the static pressure field. The pressure drop across the heatsink fin array is the dominant resistance in the flow path, consistent with the fan operating point on the Delta FFB0812VH fan curve. High-pressure regions are visible at the fan inlet stagnation points, with a pressure recovery downstream of the heatsink outlet.

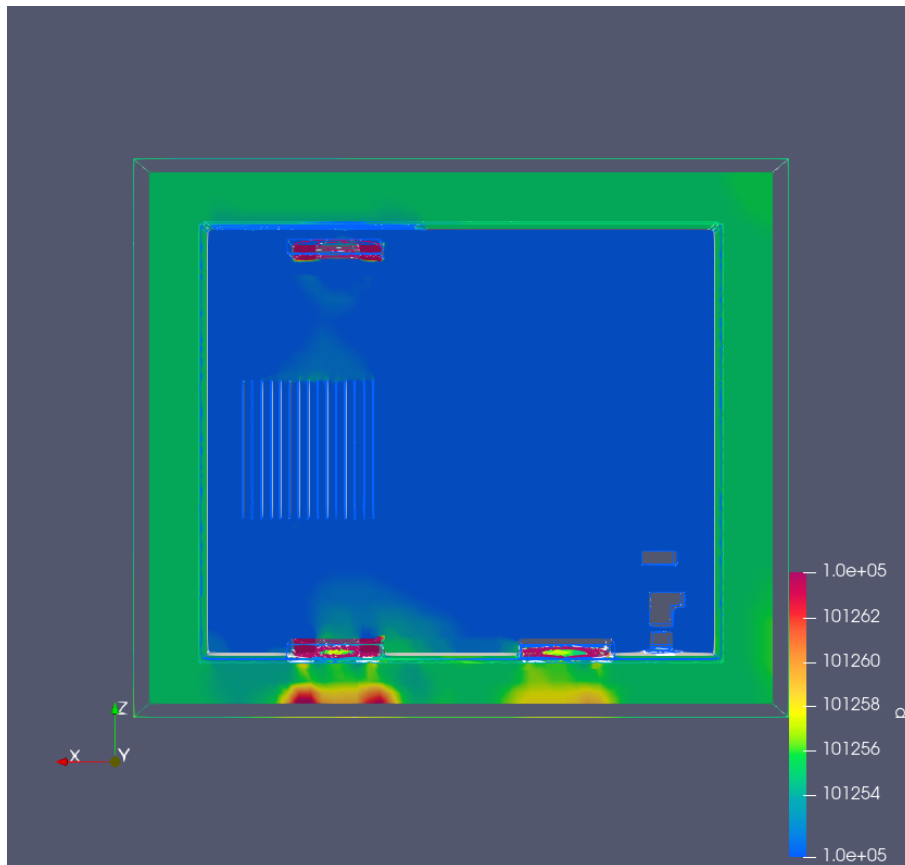


Figure 7: Static pressure contour on mid-plane slice (Pa).

5. Discussion

5.1 Thermal Performance

The predicted CPU junction temperature of 96°C under 150 W dissipation is within the typical operating envelope for server processors, though approaching thermal limits. The heatsink thermal resistance (CPU to air) can be estimated as approximately 0.46 °C/W based on $(96 - 27) / 150 = 0.46$ K/W. This is consistent with published values for comparable aluminium fin heatsinks with forced air cooling at similar flow rates.

5.2 Convergence

Full convergence was not achieved within 20,000 iterations — the CPU temperature curve shows continued slow rise at a rate of approximately 0.004°C per iteration at the final time step. This slow drift is characteristic of CHT simulations with large thermal mass ratios between solid and fluid domains. The simulation would benefit from either additional iterations or adaptive relaxation factor scheduling to accelerate convergence of the solid energy equations.

5.3 Limitations and Future Work

- The TIM layer (0.1 mm) was too thin to mesh directly and was modelled as a contact resistance boundary condition, introducing approximation in the CPU-to-heatsink thermal path.
- The outer domain boundaries use a slip condition rather than true open-atmosphere boundary conditions, which may slightly affect flow development in the inlet/outlet plenums.
- Three heatsink fins on one edge show anomalously low temperatures, likely due to insufficient mesh refinement at those fin tips — a known mesh quality limitation.
- The fan was modelled as a uniform velocity inlet rather than using the full fan curve pressure-flow relationship, which does not account for system resistance effects.
- Future work could include transient thermal analysis, radiation modelling, and parametric studies of fin pitch and fan speed.

6. Methodology Summary

Geometry: Ansys SpaceClaim (Python API V261). Multi-body STL export with FilePerBody granularity.

STL Processing: surfaceTransformPoints to scale mm → m. Manual inspection with ParaView.

Background Mesh: blockMesh with 5 mm base cell size across the full domain extents.

Surface Snapping: snappyHexMesh with castellatedMesh + snap + addLayers. locationsInMesh for multi-region detection.

Region Splitting: splitMeshRegions -cellZones to produce domain0, CPU, Heatsink, FRONT_PANEL.

Initial Flow: Isothermal run with fixedValue T=300K to establish flow field before coupling.

Coupled CHT: chtMultiRegionSimpleFoam with turbulentTemperatureCoupledBaffleMixed interfaces.

Parallel Run: 16-core MPI decomposition using scotch method across 20,000 iterations.

Post-processing: reconstructPar followed by ParaView visualisation. Python matplotlib for convergence plots.

Software Stack

Tool	Version	Purpose
OpenFOAM	v2012	CFD solver suite
Ansys SpaceClaim	2023	CAD geometry creation
ParaView	5.x	Post-processing and visualisation
Python / matplotlib	3.x	Convergence plotting
Ubuntu (WSL2)	24.04	Operating system
OpenMPI	4.0.3	Parallel communication

7. Conclusion

A steady-state conjugate heat transfer simulation of a 2U rack-mount server has been successfully completed using OpenFOAM v2012. The simulation couples turbulent forced convection in the air domain with conductive heat transfer through four solid regions, providing a comprehensive picture of the server's thermal performance under a 150 W CPU load.

Key findings include a predicted CPU junction temperature of 96°C, an estimated heatsink thermal resistance of 0.46 K/W, peak fin channel velocities of 11 m/s, and clear visualisation of the thermal plume structure downstream of the heatsink. The simulation demonstrates proficiency in the full CFD workflow from CAD geometry through meshing, solver configuration, parallel computation, and post-processing.

This work is part of an OpenFOAM portfolio demonstrating application of open-source CFD to industrially relevant thermal management problems. The full case files, including geometry scripts, mesh configuration, and solver settings, are available at github.com/samuelsimmons99/OpenFOAM-Portfolio.